Crowdsource Litter Pickup

University of CINCINNATI CEAS

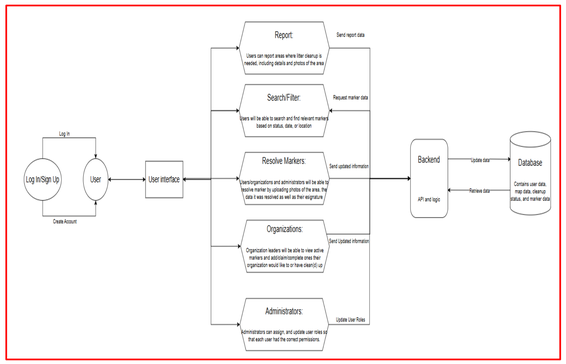
CEAS '26 EXPO

Problem & Motivation

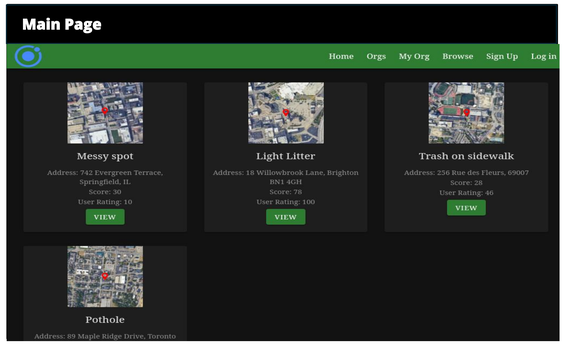
Litter accumulation poses a significant challenge to communities by degrading environmental quality, harming public health, and diminishing the appearance of shared spaces. Existing reporting systems are often fragmented or difficult to use, resulting in low participation and incomplete data. At the same time, cleanup efforts frequently lack coordination and visibility, making it difficult to prioritize high impact areas. To address these challenges, there is a need for a centralized user-friendly platform that enables real time reporting, data-driven insights, and community driven cleanup efforts in a single, cohesive system.

Project Goals

- Develop a crowdsourced platform for reporting litter in local communities
- Provide an interactive map to visualize litter locations and trends
- Enable volunteer organizations and city officials to coordinate cleanup efforts more effectively
- Improve environmental conditions and maintain cleaner public spaces



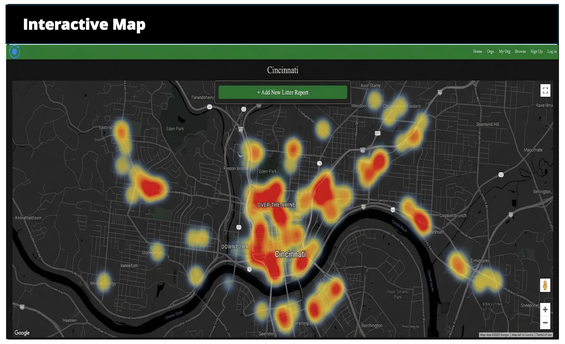
System architecture illustrating user interaction, backend processing, and data storage



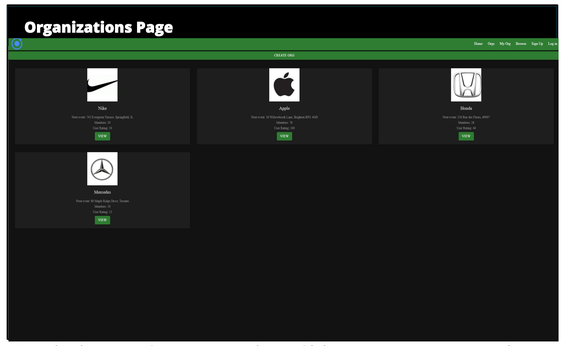
Main Page where users can browse reports

Abstract

Crowdsource Litter Pickup empowers community members to mark and report areas of heavy litter. This geotagged data is shared with volunteer groups, nonprofits, and city officials to help prioritize and coordinate cleanup efforts. By combining technology, civic participation, and sustainability, this platform promotes greater community involvement in maintaining cleaner, healthier public spaces.



View of the interactive map highlighting litter hotspots, allowing users to add new pins in areas with high concentrations of litter



Organization page where users can view and join groups to access community cleanup opportunities

System Overview

Our web application allows users to submit geotagged litter reports, including images through an Ionic React frontend. The interface integrates a map API that allows users to easily place pins and visualize reported locations in real time.

The backend is powered by Supabase, which handles data storage and user authentication. Submitted reports are securely stored and retrieved for display on the interactive map.

An AI powered classifier system built with Python and PyTorch analyzes uploaded images to estimate litter concentration and identify potential hotspots. These insights enhance the accuracy of reports and support more effective cleanup coordination by volunteers, nonprofits and municipal agencies.

Key Features

- User account creation and secure authentication
- Group creation, management, and joining
- Geotagged litter reporting with map markers
- Interactive map for visualizing and exploring submitted reports
- Image upload with AI powered analysis for litter classification
- Hotspot detection to identify areas with high litter concentration

Technology used

supabase, IONIC, PostgreSQL, Leaflet, GitHub, React, PyTorch, OpenStreetMap

Results

- Intuitive and user-friendly interface design
- Fully Functional user authentication system
- Robust database for managing users, reports and groups
- Interactive map for visualizing litter reports in real time

Challenges

- Ionic React and Supabase were new for most of the team
- Not much experience with UI design
- Real time map updates were difficult and could cause too much data to load at once
- AI model training took more time than expected
- Balancing time for the project with work and school was difficult

Broader Impacts

- Promotes cleaner and healthier communities
- Encourages civic engagement and community driven volunteering
- Supports nonprofits and local governments in coordinating and prioritizing cleanup efforts
- Enables data driven decision making through real time litter reporting and hotspot analysis
- Raises environmental awareness and fosters long term sustainability practices
- Strengthens collaboration between residents, organizations and municipalities

References

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4. https://react.dev/reference/react
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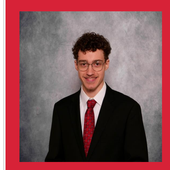

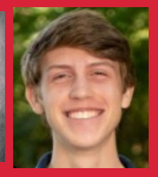

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